Substitution into Expressions and Formulae

Solutions — Short worksheet

Foundation

F1. $3x + 5$ when $x = 4$

$$3(4) + 5 = 12 + 5 = \mathbf{17}$$

F2. $2y - 3$ when $y = 8$

$$2(8) - 3 = 16 - 3 = \mathbf{13}$$

F3. $P = 22h$ when $h = 7$

$$P = 22 \times 7 = \mathbf{\$154}$$

Verify: $22/hour for 7 hours → $154. ✓

Standard

S1. $C = 50 + 35d$, find C for $d = 4$ days.

$$C = 50 + 35(4) = 50 + 140 = \mathbf{\$190}$$

Verify: 4 days at $35 = $140, plus $50 fixed = $190. ✓

S2. $4m + 3n$ when $m = -2$ and $n = 5$

$$4(-2) + 3(5) = -8 + 15 = \mathbf{7}$$

Watch the sign: 4×-2 *gives a negative.*

S3. $I = \dfrac{PRT}{100}$ when $P = 800$, $R = 6$, $T = 4$

$$I = \frac{800 \times 6 \times 4}{100} = \frac{19{,}200}{100} = \mathbf{\$192}$$

Verify: 6% of $800 is $48 per year. Over 4 years: $4 \times 48 = \$192$. ✓

Higher

H1. $2a^2 - 3ab$ when $a = -1$ and $b = 4$

$$2(-1)^2 - 3(-1)(4) = 2(1) - (-12) = 2 + 12 = \mathbf{14}$$

Two sign traps: $(-1)^2 = +1$, *and subtracting* -12 *becomes* $+12$.

H2. $A = P(1 + r)^n$ when $P = 2000$, $r = 0.04$, $n = 3$

$$\begin{aligned} A &= 2000 \times (1 + 0.04)^3 \\ &= 2000 \times (1.04)^3 \\ &= 2000 \times 1.124864 \\ &= 2249.728 \\ &\approx \mathbf{\$2249.73} \quad \text{(rounded to nearest cent)} \end{aligned}$$

Verify year-by-year: $\$2000 \to \$2080 \to \$2163.20 \to \$2249.73.$ ✓

Exit ticket. $5x - 2$ when $x = 3$

$$5(3) - 2 = 15 - 2 = \mathbf{13}$$

Common student errors to watch for:

F1, F2, exit ticket: Writing $3x = 34$ for "$3x$ when $x = 4$" (concatenation instead of multiplication).

F3, S1, S3: Forgetting to include the units or the dollar sign in the answer in context.

S2: Getting -23 or 23 by mishandling the sign on the $4m$ term. The correct working is $-8 + 15 = +7$.

H1: Two common errors: (1) computing $(-1)^2 = -1$ (it equals $+1$); (2) computing $2 - 12 = -10$ instead of $2 - (-12) = 14$.

H2: Computing $(1 + 0.04)^3$ as $1 + 0.04^3$ or 1.04×3. Reinforce that $(1.04)^3 = 1.04 \times 1.04 \times 1.04$.